

Supplementary Materials for
Photonic arbitrary Chern vectors

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Supplementary Text
Figs. S1 to S7

Supplementary Text

1. Three-dimensional (3D) Haldane model

Fig. S1A show the unit cell of the 3D Haldane model, whose Bloch Hamiltonian is given by $H(k) = \begin{pmatrix} \Delta^+ & \Phi^- \\ \Phi^+ & \Delta^- \end{pmatrix}$, where $\Delta^\pm = \pm M + 2t_{a,b} \cos k_3 + \theta^\pm(t_2, \phi)$ with $\theta^\pm(t_2, \xi) = 2t[\cos(\xi \pm k_1) + \cos(\xi \mp k_2) + \cos(\xi \mp k_1 \mp k_2)]$ and $\Phi^\pm = t_1[1 + e^{\pm ik_1} + e^{\pm ik_2}]$. The 3D Haldane model yields a nontrivial bandgap (green region) ranging from -0.77 to 0.77, as shown in Fig. S1B. The corresponding parameters are set as $M = 0$, $t_1 = 1$, $t_2 = -0.2$, $t_a = 0.2$, $t_b = -0.2$, and $\phi = -\pi/2$. We characterize the topological invariant of the bandgap using the Wilson loop approach (Fig. S1C), revealing that the Chern vector is $\mathbf{C} = (0, 0, -1)$ with only a single nonzero component. Figs. S1 (D to I) shows the calculated energy-dependent surface dispersions for the $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively. We found that the chiral surface states are present on the $(0\bar{1}0)$ and $(\bar{1}00)$ side surfaces, while no such states are observed on the (001) surface. This confirms that the chiral surface states reside only on the four side surfaces, rather than covering all six surfaces. Furthermore, the chiral surface states exhibit chiral transport along the k_1 and k_2 directions, whereas the transport along the k_3 directions is nonchiral, leading to robust one-way propagation only within the direction perpendicular to the Chern vector. Then we calculate the surface dispersion contours at 0.5, as shown in Figs. S1 (J to O). The Fermi loops do not exhibit winding around the surface of the Brillouin zone (BZ).

2. Tilted 3D Haldane model

Fig. S2A shows the unit cell of the tilted 3D Haldane model. Fig. S2B presents the bulk band structure of the tilted 3D Haldane model using the same parameters as the 3D Haldane model. There is a complete 3D energy gap (green region) spanning -0.64 to 0.64. The Wannier centers of the first band (Fig. S2C) reveal that the Chern vector of the bandgap is $\mathbf{C} = (1, 1, -1)$. Figs. S2 (D to I) shows the calculated energy-dependent surface dispersions for the $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively. It can be seen that, for each surface, the surface-state dispersions exhibit positive or negative group velocities along two distinct wave-vector directions, indicating the existence of chiral surface states propagating along the diagonal direction on all surfaces. Then we calculate the surface dispersion contours at 0.5, as shown in Figs. S2 (J to O). The Fermi loops

wind around both the median and longitude of the $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surface Brillouin zone (BZ), forming torus knots of type $T(1, -1)$, $T(1, -1)$, and $T(1, 1)$, respectively.

3. Calculation methods of the topological Chern vector

A 3D Chern insulator is characterized by a topological Chern vector consisting of three Chern numbers $\mathbf{C} = (C_1, C_2, C_3)$. These three Chern numbers can be computed by slicing the 3D BZ (a three-torus T^3 parameterized by $(k_1, k_2, k_3) \in [0, 2\pi]^3$) to two-dimensional (2D) BZs (a two-torus T^2) on three different orthogonal momentum-space planes. The Chern number on each 2D BZ is given by

$$C_{\mu\nu}(k_\varepsilon) = (1/2\pi) \iint_{T^2} \Omega_{\mu\nu} dk_\mu dk_\nu \quad (1)$$

Where μ, ν , and ε represent three momentum dimensions and $\Omega_{\mu\nu}(\mathbf{k}) = \partial_{k_\mu} \mathbf{A}_\nu(\mathbf{k}) - \partial_{k_\nu} \mathbf{A}_\mu(\mathbf{k})$ is the Berry connection, where $\mathbf{A}(\mathbf{k}) = i\langle \mathbf{u}(\mathbf{k}) | \nabla_{\mathbf{k}} | \mathbf{u}(\mathbf{k}) \rangle$ is the Berry connection, and $\mathbf{u}(\mathbf{r})$ is the periodic part of the Bloch eigenstates. As a result, the Chern vector on each 2D BZ can be rewritten as $\mathbf{C}_{T^2} = (C_{k_2 k_3}(k_1), C_{k_3 k_1}(k_2), C_{k_1 k_2}(k_3))$.

Here, we take $C_{k_3 k_1}(k_2)$ for example. $C_{k_3 k_1}(k_2)$ is defined in the 2D BZ at $k_3 - k_1$ plane for each fixed k_2 . In our case, we can choose the rectangular 2D BZ, as shown in Fig. S3A, which is defined by the reciprocal lattice vectors discretizing the unit cell in the k_1 and k_3 directions. Here, we adopt the Wilson loop approach to calculate $C_{k_3 k_1}(k_2)$

$$W(k_3) = e^{-i\theta(k_3)} = \prod_{k_2} \langle u_{(k_3, k_1)}(\mathbf{r}) | u_{(k_3, k_1 + \Delta k_1)}(\mathbf{r}) \rangle, \quad (2)$$

The Wannier center along the loop k_1 for a fixed k_3 is given by $\theta(k_3) = -\text{Im}[\ln W(k_3)]$. In numerical calculations, $C_{k_3 k_1}(k_2)$ is obtained by counting the winding number of $\theta(k_3)$ as k_3 changes from 0 to 2π . We first compute the Wannier center at $k_2 = 0$ and $k_2 = \pi/a_2$, as shown in Figs. S3 (B and C). The Wannier center winds from $-\pi$ to $+\pi$ one times with a positive slope, revealing that $C_{k_3 k_1}(k_2)$ is -1. We then calculate $C_{k_3 k_1}(k_2)$ for all k_2 (from $-\pi$ to $+\pi$), as shown in Fig. S3D. It can be seen that $C_{k_3 k_1}(k_2)$ remains unchanged with the varying of k_2 , meaning that the C_2 component is 1. Similarly, $C_{k_2 k_3}(k_1) = 1$ and $C_{k_2 k_1}(k_3) = -1$ lead to C_1 and C_3 components are 1 and -1, as shown in Figs. S3 (E to H) and Figs. S3 (I to L), respectively.

4. The iso-frequency contour analysis of the nonreciprocal and reflectionless topological positive and negative refractions at the hinges separating different surfaces.

As shown in Fig. S4, nonreciprocal and reflectionless topological positive (negative) refractions occur at the six red (blue) hinges separating different surfaces. To interpret this unique phenomenon, we first apply the phase-matching conditions at one of the red hinges between the $(0\bar{1}0)$ and (100) surfaces. As shown in Fig. S4A, the refracted wavevector t retains the same direction as the incident wavevector i without the reflected wavevector r , indicating reflectionless topological positive refraction occurs at the red hinges. This is because the iso-frequency contour of the (100) surfaces contains the same wavevectors k as the $(0\bar{1}0)$ surfaces. However, the reflected wavevector r is forbidden because the iso-frequency contour of the (100) surfaces does not exist a wavevector k that satisfies the phase-matching condition. As such, there would be no reflection at the hinges, and the chiral surface states on the $(0\bar{1}0)$ surface will be totally positively refracted into the (100) surface. Then we apply phase-matching conditions to one of the blue hinges separating the $(\bar{1}00)$ and $(0\bar{1}0)$ surfaces, as shown in Fig. S4B. In order to satisfy the phase-matching condition, the direction of the refracted wavevector t needs to be mirrored to that of the incident wavevector i about the hinge, and the reflected wavevector r is forbidden, indicating that reflectionless topological negative refraction occurs at the blue hinges.

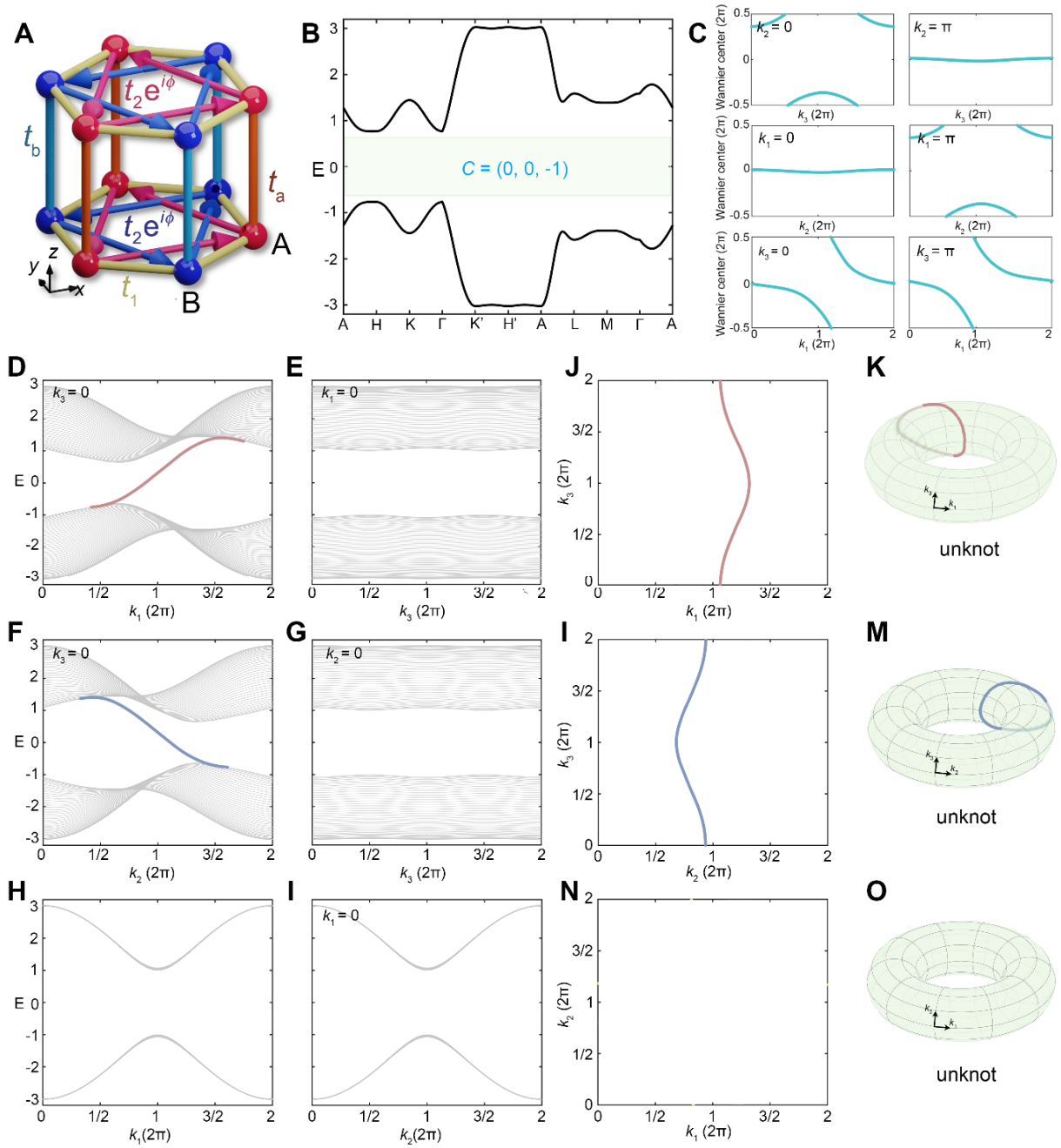


Fig. S1. 3D Haldane model with a Chern vector $\mathbf{C} = (0, 0, -1)$. (A) Schematic of the 3D Haldane model. (B) Calculated bulk band structure of the 3D Haldane model, where the green region represents the 3D complete energy gap. (C) The Wannier centers lie at k_1 - k_3 , k_2 - k_3 , and k_1 - k_2 planes, respectively, revealing that the Chern vector is $\mathbf{C} = (0, 0, -1)$. (D to I) Calculated energy-dependent surface dispersions of the chiral surface states on $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively. (J to O) Calculated surface dispersion contours (left panels) and their corresponding torus knots (right panels) formed by winding the Fermi loops on $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively.

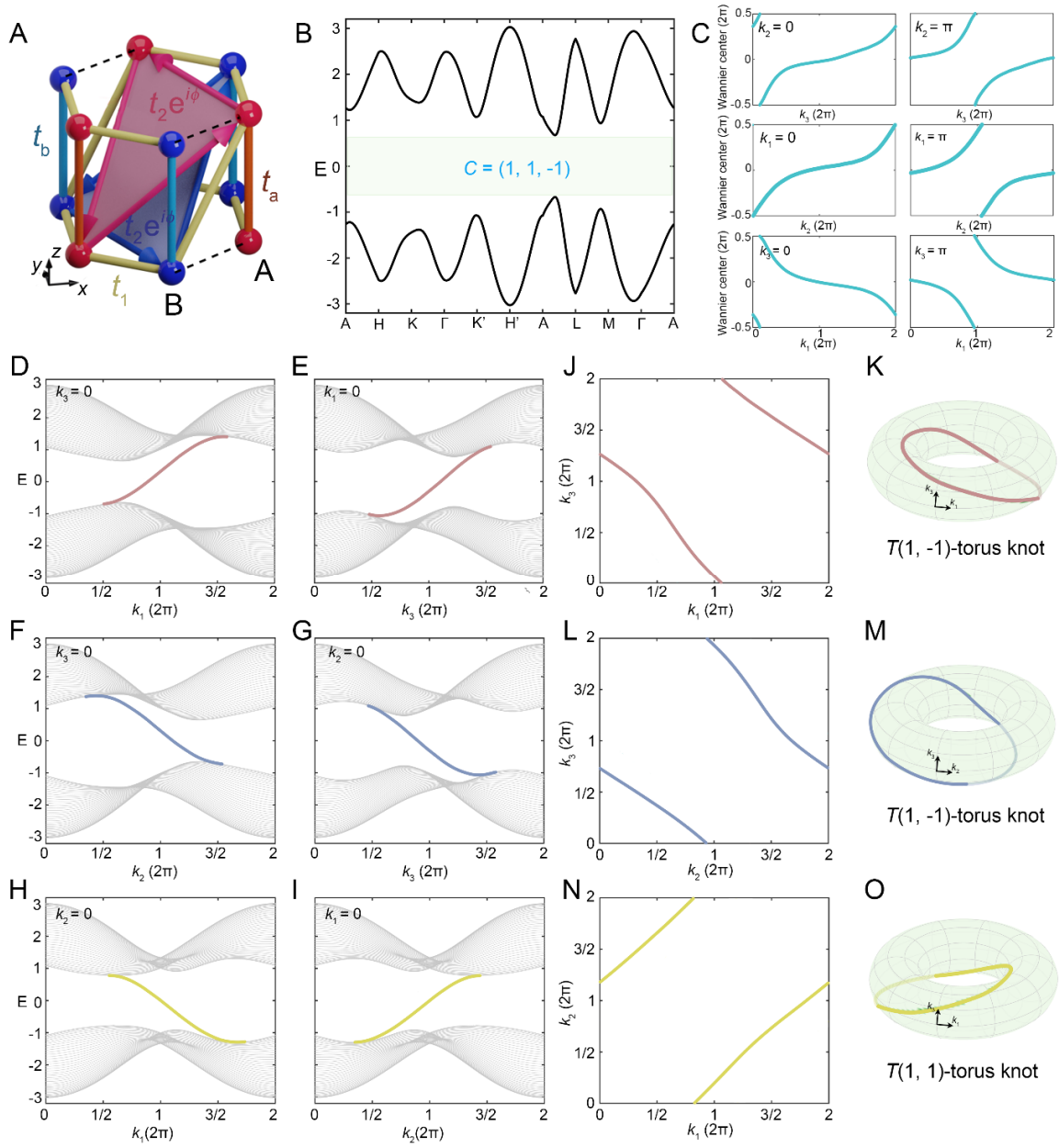


Fig. S2. Tilted 3D Haldane model with a Chern vector $\mathbf{C} = (1, 1, -1)$. (A) Schematic of the tilted 3D Haldane model. (B) Calculated bulk band structure of the tilted 3D Haldane model, where the green region represents the 3D complete energy gap. (C) The Wannier center at k_1 - k_3 , k_2 - k_3 , and k_1 - k_2 plane, respectively, revealing that the Chern vector is $\mathbf{C} = (1, 1, -1)$. (D to I) Calculated energy-dependent surface dispersions of the chiral surface states on $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively. (J to O) Calculated surface dispersion contours (left panels) and their corresponding torus knots (right panels) formed by winding the Fermi loops on $(0\bar{1}0)$, $(\bar{1}00)$, and (001) surfaces, respectively.

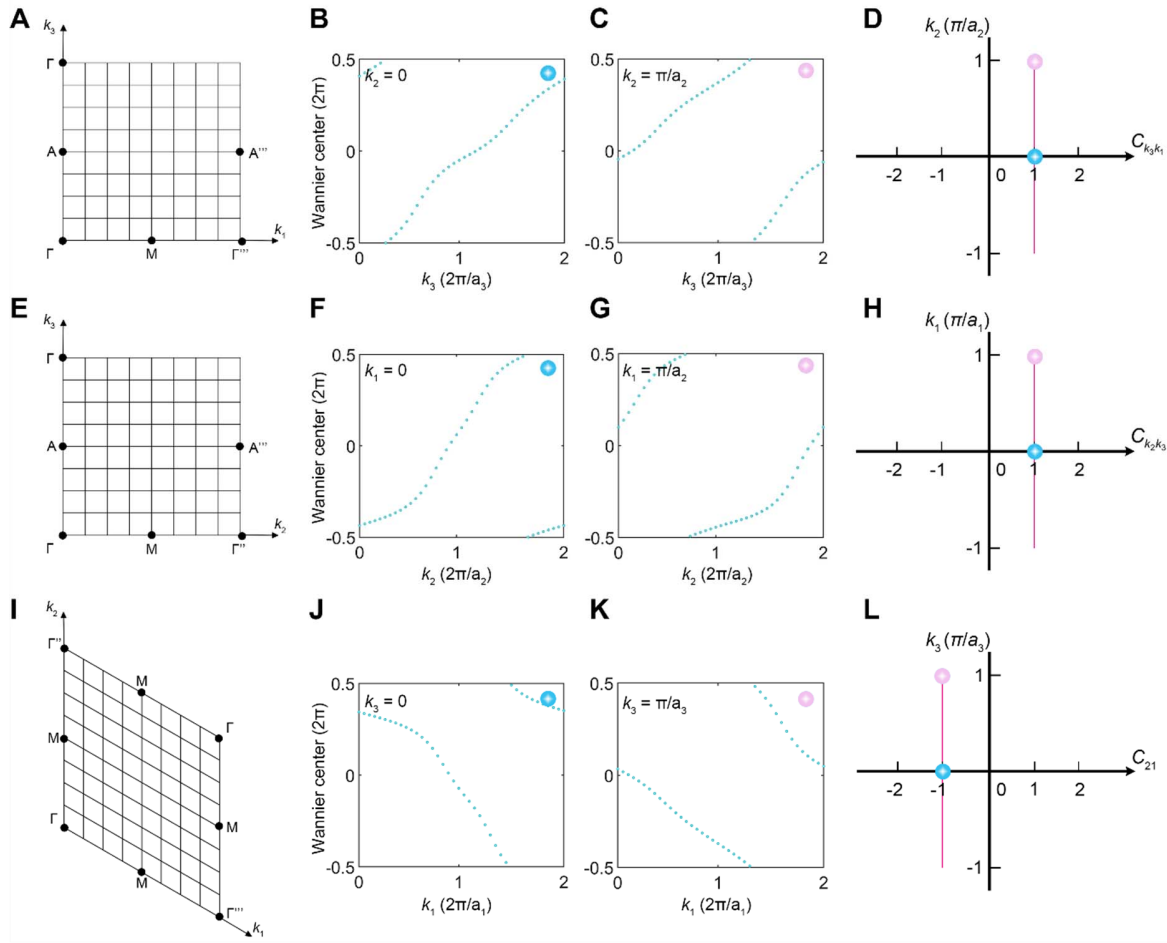


Fig. S3. Calculation of the topological Chern vector in a 3D gyromagnetic photonic crystal. (A) Discretization of the 2D BZ (k_1 - k_3 plane). (B and C) Evolution of the Wannier centers in k_1 - k_3 plane at $k_2 = 0$ and $k_2 = \pi/a_2$, revealing that $C_{k_3k_1}(k_2 = 0)$ and $C_{k_3k_1}(k_2 = \frac{\pi}{a_2})$ are 1. (D) $C_{k_3k_1}(k_2)$ as a function of k_2 , indicating the Chern vector component $C_2 = 1$. (E to H) The same as (A to D) but for $C_{k_2k_3}$. (I to J), The same as (A to D) but for $C_{k_2k_1}$.

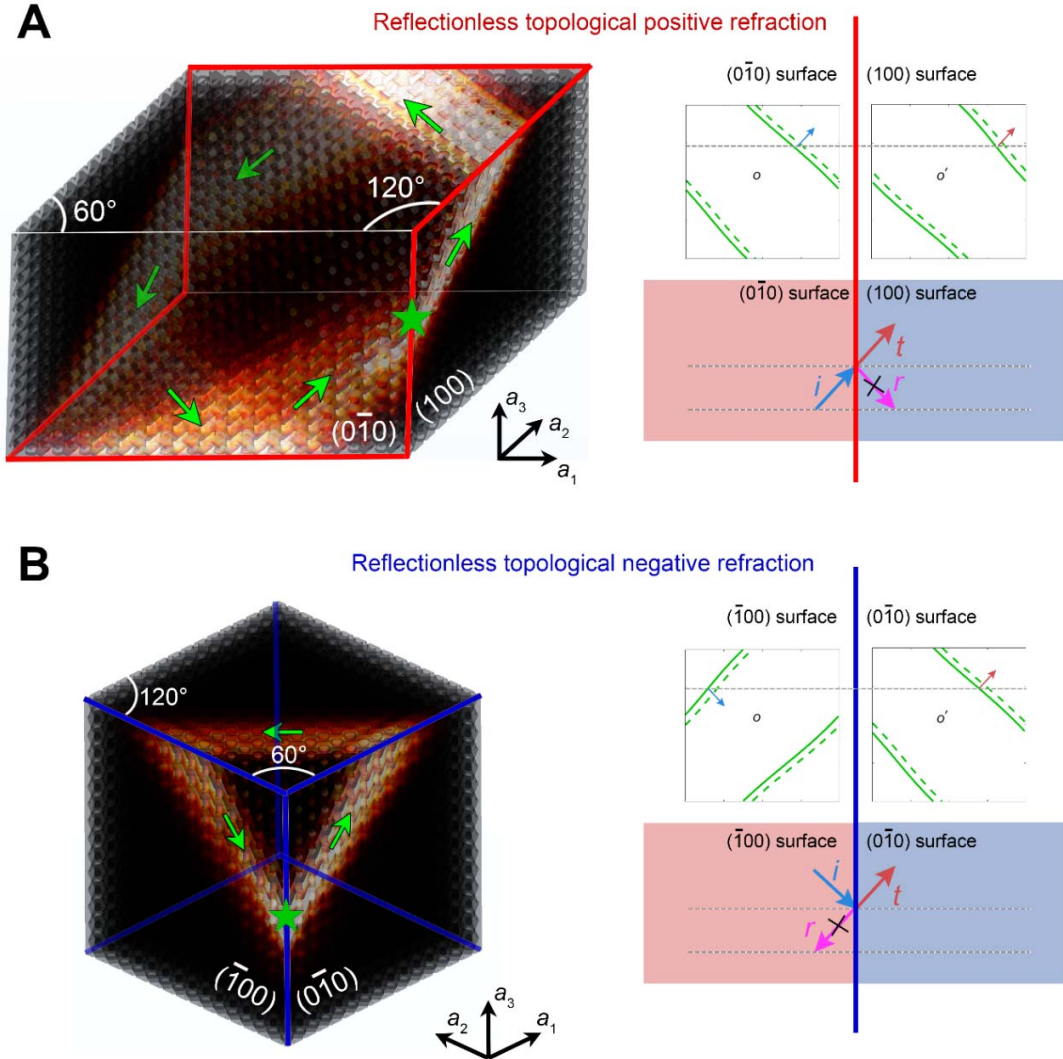


Fig. S4. The iso-frequency contour analysis of the nonreciprocal topological positive and negative refractions at the hinges separating different surfaces. (A) Left panel: simulated electric field distribution of reflectionless topological positive refraction of the chiral surface states at the red hinges, the green star represents the point source, and the green arrows indicate the propagation path of the chiral surface states. Right panel: iso-frequency contour analysis at the red hinges separating the $(0\bar{1}0)$ and (100) surfaces. The solid and dashed green lines denote the iso-frequency contours at 14.3 GHz and at a slightly higher frequency, respectively. Colored arrows labeled by i , t , and r represent the wave vectors of incident, refracted, and reflected chiral surface states, respectively. **(B)** The same as (A) but for the reflectionless topological negative refraction of the chiral surface states at the blue hinges. The simulated electric field distributions in (A) and (B) are reproduced from Fig. 2B and Fig. 2D, respectively, to maintain a clear link between the main text and the Supplementary Materials.

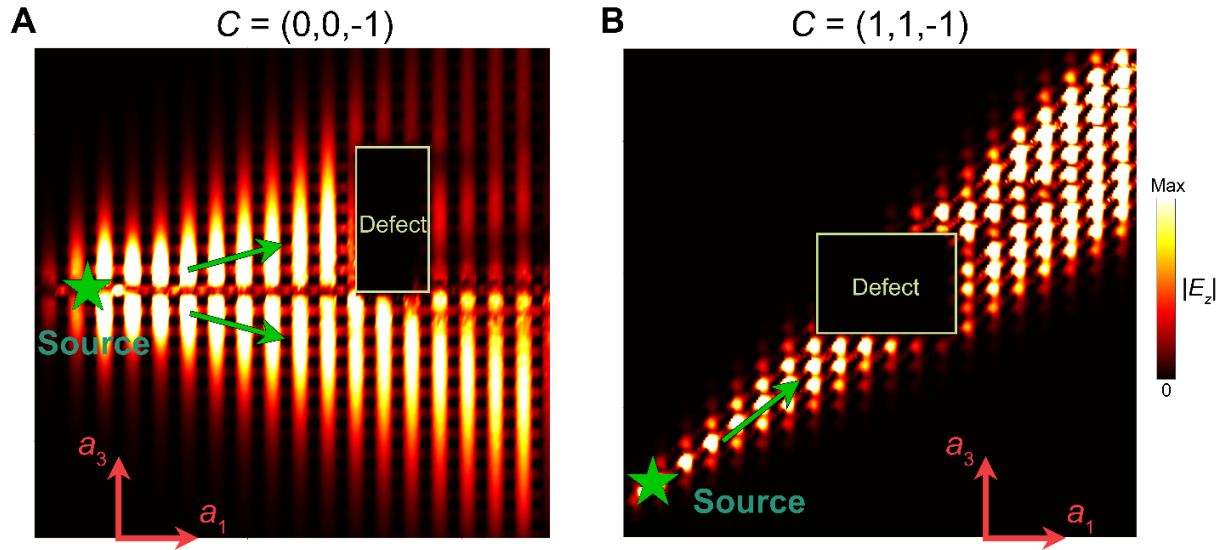


Fig. S5. Comparison of the robustness of the chiral surface states in 3D photonic Chern insulators with Chern vectors $C = (0, 0, -1)$ and $C = (1, 1, -1)$. (A and B) Simulated electric field distributions of the chiral surface states at 14.3 GHz with a metallic defect on the $(0\bar{1}0)$ surface of 3D photonic Chern insulators possessing Chern vectors $C = (0, 0, -1)$ (A) and $C = (1, 1, -1)$ (B), revealing the genuine chiral surface states with enhanced robustness in 3D photonic Chern insulators with arbitrary Chern vectors.

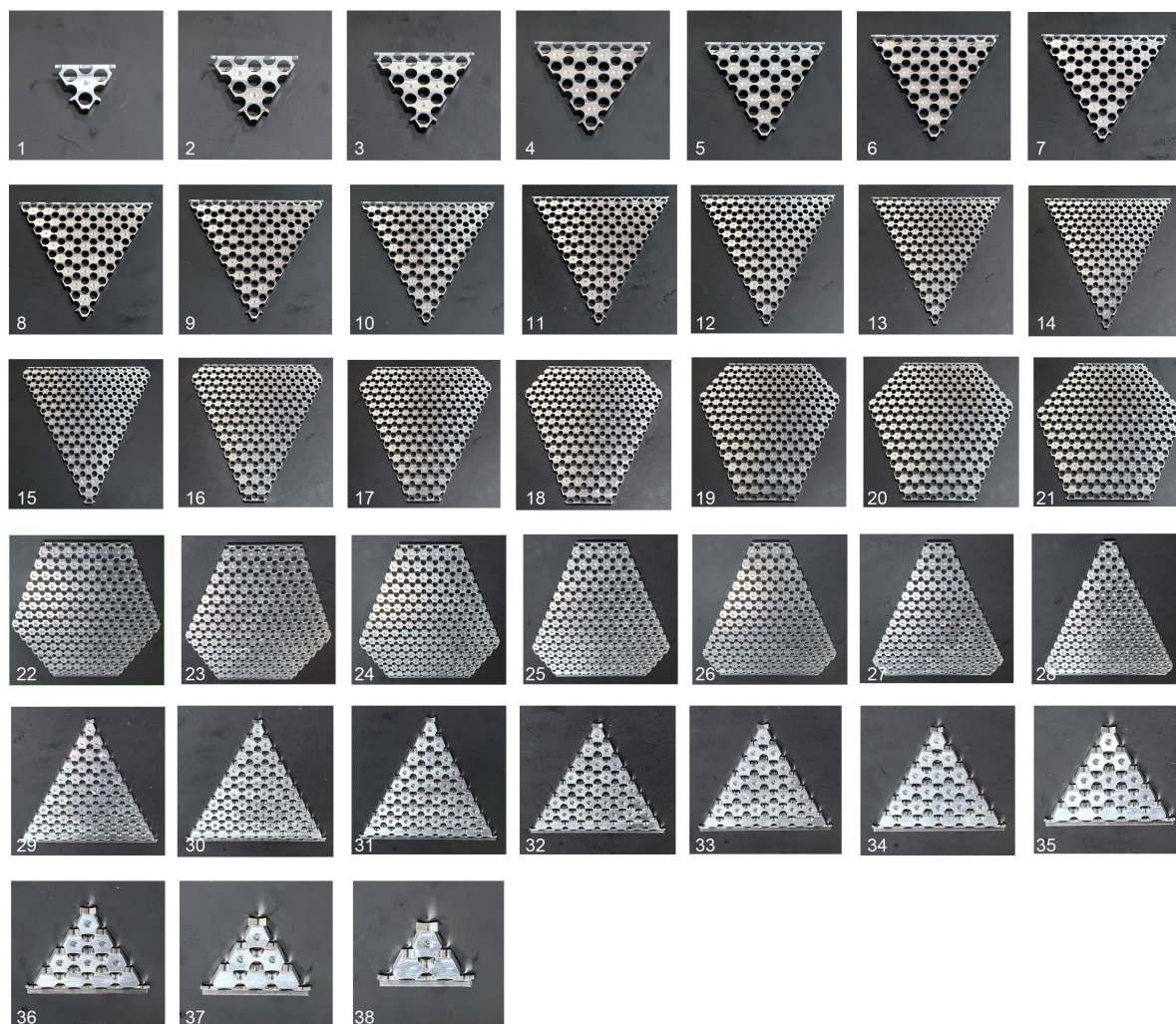


Fig. S6. Experimental sample details. The photograph of 38 layers of perforated metallic plates.

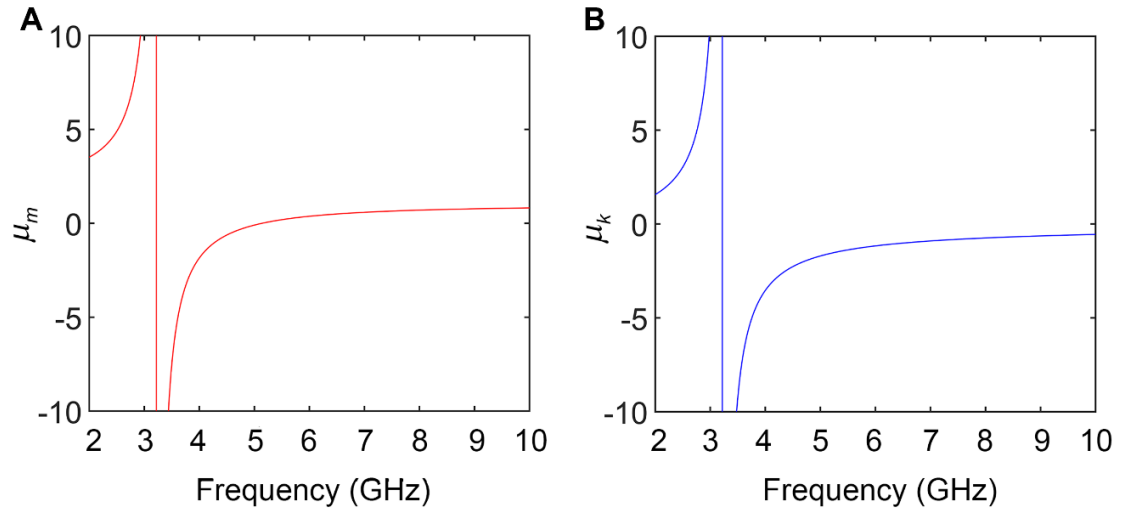


Fig. S7. Permeability tensor of the gyromagnetic material. (A and B) Frequency-dependent elements μ_m and μ_k of the permeability tensor of the gyromagnetic material for $\mu_0 H_0 = 0.26$ T.